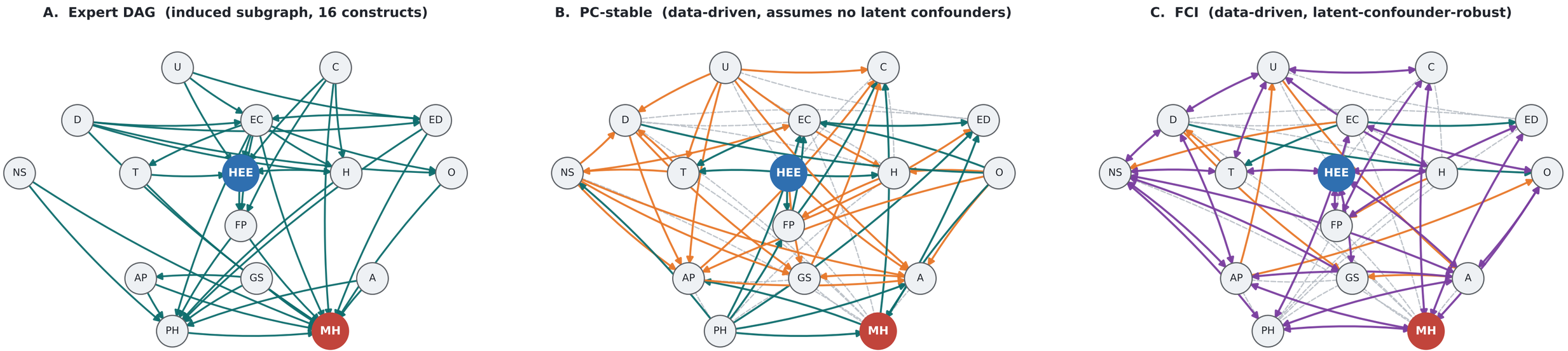
Expert-elicited DAG vs constraint-based causal discovery (HEE–MH structure, LSOA cross-section)



Shared node layout. PC recovers 19/40 expert adjacencies (recall 0.47, precision 0.42); FCI recovers 16/40 (recall 0.40, precision 0.41) and marks 28/39 edges bidirected (↔, latent common cause) — including all 5 HEE edges. HEE and MH are non-adjacent in all three graphs — the DAG's fully-mediated HEE→MH claim, recovered with and without a causal-sufficiency assumption. Orientations are CPDAG/PAG equivalence-class statements; all associations are area-level.

Directed adjacency shared with expert DAG — Discovered-only (novel) — Expert edge not recovered — FCI bidirected ↔ (latent common cause) — HEE (exposure) — MH (outcome)

A = Accessibility AP = Air Pollution C = Climate D = Demographics EC = Economic Capital ED = Education FP = Fuel Poverty GS = Green Space
H = Housing Characteristics HEE = Home Energy Efficiency MH = Mental Health NS = Neighbourhood Stress O = Overcrowding PH = Physical Health T = Tenure U = Urbanicity